

Civil Engineering B.Tech - 2023 onwards

Semester I				
Course Code	Course	Credit	Segment	Type
MA1110	Calculus I	1	xx	BS
MA1220	Calculus II	1	xx	BS
EP1108	Modern Physics	2	xx	BS
CY1010	Environmental Chemistry	2	xx	BS
BT1010	Introduction to Life Sciences	1	xx	BS
ID1063	Introduction to Programming	3	16	BE
ID1171	Fabrication Lab-I	2	16	BE
ID1041	Engineering Drawing	2	16	DC
LA1760	English Communication	2	xx	LA

Total **16**
 Basic Science 7
 Basic Engg 5
 Dept. Core 2
 LA 2

Semester II				
Course Code	Course	Credit	Segment	Type
MA1140	Elementary Linear Algebra	1	12	BS
MA1150	Differential Equations	1	34	BS
CY1031	Chemistry Laboratory	2	14	BS
ME2120	Thermodynamics	3	16	BE
EE1102	Basic Electrical Engineering	3	16	BE
ID1054	Digital Fabrication	2	16	BE
ME1020	Engineering Mechanics	3	16	BE
CE1100	Construction Materials	3	16	DC
LA XXXX	Liberal/Creative Arts	1	XX	LA

Total **19**
 Basic Science 4
 Basic Engg 11
 Dept. Core 3
 LA/CA 1

Semester III				
Course Code	Course	Credit	Segment	Type
MA2110	Introduction to Probability	1	12	BS

MA2120	Transform Techniques	1	34	BS
CE2420	Fluid Mechanics	3	16	DC
CE2130	Mechanics of Solids and Structures	3	16	DC
CE2510	Environmental Systems Engineering I	3	16	DC
CE2121	Construction Materials Lab	2	16	DC
CE2920	Surveying	3	16	DC
CE2900	Construction Management	2	14	DC
CE2600	Introduction to Sustainability in Civil Engineering	1	56	DC
Total		19		
Basic Science		2		
Dept. Core		17		

Semester IV				
Course Code	Course	Credit	Segment	Type
MA2140	Introduction to Statistics	1	12	BS
MA2130	Complex Variables	1	12	BS
CE2111	Structural Mechanics Lab	2	36	DC
CE2140	Structural Analysis	3	16	DC
CE2300	Geotechnical Engineering	3	16	DC
CE2301	Geotechnical Engineering Lab	2	16	DC
CE2520	Environmental Systems Engineering II	3	16	DC
CE2410	Hydraulic Engineering	3	16	DC
Total		18		
Basic Science		2		
Dept. Core		16		

Semester V				
Course Code	Course	Credit	Segment	Type
CE3330	Foundation Engineering	3	16	DC
CE3212	Reinforced Concrete Design	3	16	DC
CE3400	Engineering Hydrology	2	14	DC
CE3401	Hydraulic Engineering Lab	1	56	DC
CE3511	Environmental Engineering lab	2	36	DC
CE3411	Fluid Mechanics Lab	1	14	DC
CE3242	Structural Steel Design	3	16	DC
CE3820	Highway Design and Materials	2	14	DC
CE3821	Highway Materials Lab	1	56	DC

Total **18**
Dept. Core 18

Semester VI				
Course Code	Course	Credit	Segment	Type
CE XXXX/CE4057/CE 2025	Dept. Electives/Internship*/Project	6	16	DE
XX XXXX	Free Electives	6	16	FE
LA XXXX	Liberal/Creative Arts	2	XX	LA
ID1050	Artificial Intelligence	1	XX	BE

Total **15**
Dept. Elective 6
Free Electives 6
Basic Engg 1
LA/CA 2

Semester VII				
Course Code	Course	Credit	Segment	Type
CE XXXX/CE4025	Dept. elective/Project	3	16	DE
CE4400	Water Resources Engineering	2	16	DC
CE4840	Traffic Engineering and Planning	2	14	DC
CE4841	Traffic Engineering Lab	2	36	DC
CE4830	Railway and Airport Engineering	1	56	DC
LA XXXX	Liberal/Creative Arts	2	XX	LA

Total **12**
Dept. Elective 3
Dept. Core 7
LA/CA 2

Semester VIII				
Course Code	Course	Credit	Segment	Type
XX XXXX	Free Electives	7	XX	FE
XX XXXX	Ethics and Values	1	XX	LA
LA XXXX	Liberal/Creative Arts	2	XX	LA

Total **10**
Free electives 7
LA/CA 3

Summary		
Type	Credits	Permissible Limits as per guidelines
Basic Science (BS)	15	12%-13%
Basic Engineering (BE)	17	12%-13%
LA/CA	10	7%-8%
Dept. Core + Dept. Elective (DC+DE)	72	55%-60%
Free electives (FE)	13	10%
Total Credits	127	Total credits 125-130.
% of 3 credit courses including free electives	>40%	

SGuidelines for taking Departmental Electives/Projects
a) Total credits for Departmental Electives/Projects should be minimum 9 credits
b) Core electives can be 1, 2, or 3 credit courses
c) If a project is to be taken, it should be of 3 credits.
d) If one is fulfilling total of nine credits by taking Project of three credits and Core electives of six credits, three credits of core electives are to be taken from Basket 1 and remaining credits of core electives are to be taken from Basket 2.
e) If one is fulfilling total of nine credits by taking only Core electives, at least six credits of Core electives are to be taken from Basket 1.
Guidelines for taking LA/CA Courses
a) Maximum 40% credits of CA courses.
b) Minimum 60% credits of LA courses

Basket 1 (Departmental Electives)			
Course Code	Course name	Credits	Prerequisite
CE5200	Finite element analysis	3	CE2140
			MA1140
			CE5110
			CE5120
CE6222	Prestressed concrete Design	3	CE3212
			CE5212
CE5290	Applied elasticity and Plasticity	3	CE2140
			MA1150
			MA1140
			CE5110
CE5212	Advanced reinforced concrete design	3	CE2140
CE5140	Structural dynamics	3	MA2120
			MA1150
			MA1140
			CE5110
CE5250	Advanced structural mechanics	3	CE2140
			MA1150
			MA1140
			CE2140
CE5232	Advanced structural steel design	3	CE3242
CE5150	Stability of Structures	3	CE3242
CE5224	Structural Condition Assessment and Strengthening	3	CE5212
CE5270	Design of Cold-Formed Light Gauge Steel Structures	3	CE3242
CE5170	Mathematical Methods in Civil Engineering	3	
CE5900	Machine Learning and Scientific Computing for Civil Engineers	3	
CE5190	Advanced Concrete Technology	3	
CE5310	Advanced Soil Mechanics	3	
CE5330	Soil Dynamics	3	
CE5300	Advanced Foundation Engineering	3	
CE5340	Ground Modification Techniques	3	
CE5320	Introduction to Railway Engineering	1	
CE5352	Design of Earth Structures	3	
CE6302	Design Studio	3	
CE6323	Experimental Soil Mechanics	3	
CE6350	Railway Geotechnics	3	
CE6360	Geotechnical Earthquake Engineering	3	
CE6370	Soil-Structure Interaction	3	
CE6390	Pavement Geotechnics	3	
CE6392	Designing with Geosynthetics	3	
CE6410	In-situ Testing	3	
CE5730	Urban Transportation Systems Planning	3	
CE6740	Analysis and Design of Pavements	3	CE3820
CE5220	Solid Waste Management	3	
CE5210	Biochemical Processes in Water and Wastewater Engineering	3	
CE6143	Life Cycle Assessment & Sustainability	2	
CE6540	Contaminant Hydrology and Remediation	3	
CE6210	Physicochemical Processes in Water and Wastewater Engineering	3	
CE6500	Engineering Hydrology and Hydrologic Systems	3	
CE6520	Irrigation Water Management	2	
CE5510	Open-Channel Hydraulics and Sediment Transport	3	
CE5530	Groundwater Modelling	3	
CE6501	Applied Computational Laboratory	2	
CE6630	Water Quality Modeling	3	
CE6511	Hydraulic and Hydrologic Simulation Lab	2	
CE3420	Fundamentals of GIS and Remote Sensing	2	
CE3421	RS and GIS Lab	1	
CE6610	Remote Sensing & GIS Applications in Water Resources	3	
CE6670	Advanced Statistical Methods in Hydrology	3	CE6501

Basket 2 : Institute-wide electives		
Course Code	Course name	Credits
CE5200	Finite element analysis	3

CE6222	Prestressed concrete Design	3
CE5290	Applied elasticity and Plasticity	3
CE5212	Advanced Reinforced Concrete Design	3
CE5140	Structural dynamics	3
CE5250	Advanced structural mechanics	3
CE5232	Advanced steel design	3
CE5150	Stability of Structures	3
CE5224	Structural Condition Assessment and Strengthening	3
CE5270	Design of Cold-Formed Light Gauge Steel Structures	3
CE5170	Mathematical Methods in Civil Engineering	3
CE5190	Advanced Concrete Technology	3
CE5900	Machine Learning and Scientific Computing for Civil Engineers	3
CE5310	Advanced Soil Mechanics	3
CE5330	Soil Dynamics	3
CE5300	Advanced Foundation Engineering	3
CE5340	Ground Modification Techniques	3
CE5320	Introduction to Railway Engineering	1
CE5352	Design of Earth Structures	3
CE6302	Design Studio	3
CE6323	Experimental Soil Mechanics	3
CE6350	Railway Geotechnics	3
CE6360	Geotechnical Earthquake Engineering	3
CE6370	Soil-Structure Interaction	3
CE6390	Pavement Geotechnics	3
CE6392	Designing with Geosynthetics	3
CE6410	In-situ Testing	3
CE5730	Urban Transportation Systems Planning	3
CE6740	Analysis and Design of Pavements	3
CE5220	Solid Waste Management	3
CE5210	Biochemical Processes in Water and Wastewater Engineering	3
CE6143	Life Cycle Assessment & Sustainability	2
CE6540	Contaminant Hydrology and Remediation	3
CE6210	Physicochemical Processes in Water and Wastewater Engineering	3
CE6500	Engineering Hydrology and Hydrologic Systems	3
CE6520	Irrigation Water Management	2
CE5510	Open-Channel Hydraulics and Sediment Transport	3
CE5530	Groundwater Modelling	3
CE6501	Applied Computational Laboratory	2
CE6630	Water Quality Modeling	3
CE6511	Hydraulic and Hydrologic Simulation Lab	2
CE3420	Fundamentals of GIS and Remote Sensing	2
CE3421	RS and GIS Lab	1
CE6610	Remote Sensing & GIS Applications in Water Resources	3
CE6670	Advanced Statistical Methods in Hydrology	3
ME2220	Kinematics and Dynamics of Mechanisms	3
ME3413	Machine drawing and solid modeling	2
ME2070	Introduction to mathematical modeling	1
ME5010	Mathematical methods for engineers	3
ME5110	Advanced mechanics of solids	3
ME5120	Dynamics and vibration	3
ME5700	Analysis and design of composite structures	3
ME5610	Fracture mechanics	3
ME5630	Nonlinear oscillation	3
ME5690	Advanced FEM	3
ME5260	Continuum mechanics	3
ME5320	Advanced heat transfer	3
ME5330	Computational fluid dynamics	3
CH5050	Computational methods for engineers	3
MSS100	Composite materials	3
MSS140	Introduction to computational methods in materials science	3